

Wildfire Smoke: breathe safe, help others

Wildfire smoke carries tiny particles that enter your lungs and bloodstream. Most harm is preventable. The two things that matter most: **stay aware of air quality** and **reduce smoke exposure indoors and out**.

MEDICAL EMERGENCY

Smoke can trigger life-threatening reactions

Call your local emergency number **immediately** if someone has chest pain, severe shortness of breath, confusion, or lips/fingernails turning blue. People with asthma or COPD should follow their emergency action plan and seek care if rescue inhalers do not help.

Air Quality Index — know the colors

The AQI tells you how dangerous the air is right now. Check **AirNow.gov** (U.S.), **Plume Labs**, or your local weather app.

0–50 GREEN

Good

Air quality is fine. Normal activities are safe for most people.

51–100 YELLOW

Moderate

Sensitive people (children, elderly, asthma/COPD) should reduce prolonged outdoor exertion.

101–150 ORANGE

Unhealthy for sensitive groups

Everyone should reduce prolonged outdoor exertion. Sensitive groups should avoid it.

151–200 RED

Unhealthy

Everyone should avoid prolonged exertion outdoors. Sensitive groups should stay inside.

201–300 PURPLE

Very unhealthy

Everyone should stay inside and reduce activity. Keep windows closed.

301+ MAROON

Hazardous

Emergency conditions. Avoid all outdoor exertion. Use N95 masks if you must go out.

Who is most at risk?

- Children and teenagers (lungs still developing)

- Adults 65 and older
- Pregnant people
- People with asthma, COPD, or other lung diseases
- People with heart disease or diabetes
- Outdoor workers
- People experiencing homelessness or without reliable shelter

Protect yourself and others

- 1 **Check the AQI every morning** during wildfire season. Set alerts on AirNow or your weather app.
- 2 **Stay indoors** with windows and doors closed when AQI is Orange or worse.
- 3 **Create a clean-air room.** Choose an interior room with few windows. Run an air purifier continuously. Avoid candles, frying, vacuuming, or gas stoves without ventilation.
- 4 **Wear an N95 or P100 mask** if you must go outside when AQI is Unhealthy (Red) or worse. Cloth masks do not block PM2.5.
- 5 **Build a Corsi-Rosenthal box.** A DIY air purifier made from a box fan and 4–5 MERV-13 filters. Studies show it reduces indoor PM2.5 as well as commercial HEPA units for about \$50–\$70. See cleanaircrew.org for build guides.
- 6 **Check on vulnerable neighbors daily.** Offer masks, a ride to a clean-air shelter, or help running a filter.
- 7 **Reduce or cancel outdoor events** when AQI is over 150.

For community organizers

ACTION CHECKLIST

- Open **clean-air shelters** with HEPA filtration or Corsi-Rosenthal boxes.
- **Distribute N95 masks** to outdoor workers and unhoused neighbors.
- Share AQI alerts via text, social media, and local networks.
- Reschedule outdoor community events when AQI > 150.
- Partner with local libraries, community centers, and faith groups to extend shelter hours.

When to seek medical help

- Difficulty breathing or shortness of breath
- Persistent cough, especially with phlegm or blood
- Chest pain or tightness
- Wheezing that does not improve with medication
- Dizziness or confusion
- Palpitations or irregular heartbeat

Key resources

- **AirNow.gov** — Real-time AQI and fire maps (U.S.)
 - **Fire and Smoke Map** — fire.airnow.gov
 - **EPA Wildfire Smoke Guide** — epa.gov/pm-pollution/wildfire-smoke-guide-factsheets
 - **Corsi-Rosenthal Box build guides** — cleanaircrew.org
 - **Poison Control** — 1-800-222-1222 (24/7 U.S. medical guidance)
 - **Emergency** — 911 (U.S./Canada), 112 (Europe), 999 (UK)
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Sources: U.S. Environmental Protection Agency (EPA), Centers for Disease Control and Prevention (CDC), World Health Organization (WHO), AirNow, National Weather Service, and peer-reviewed research on DIY air filtration (Corsi & Rosenthal, 2021; Lunder et al., 2022).

This guide is for educational purposes and does not replace professional medical advice. In an emergency, call your local emergency number.